

Question 2

(30 marks)

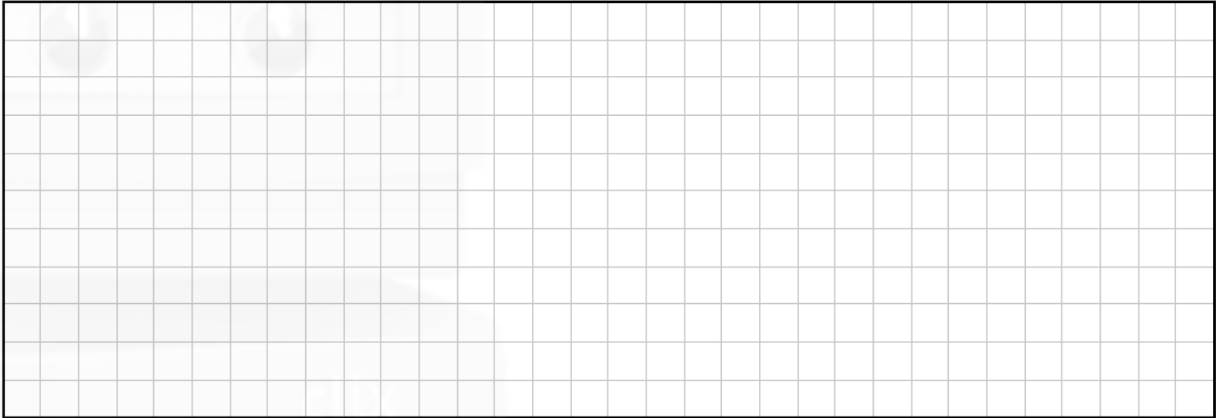
(a) At a stall in a fun fair, the probability of knocking a coconut off its support is 0.34 .

(i) What is the probability of **not** knocking a coconut off its support?

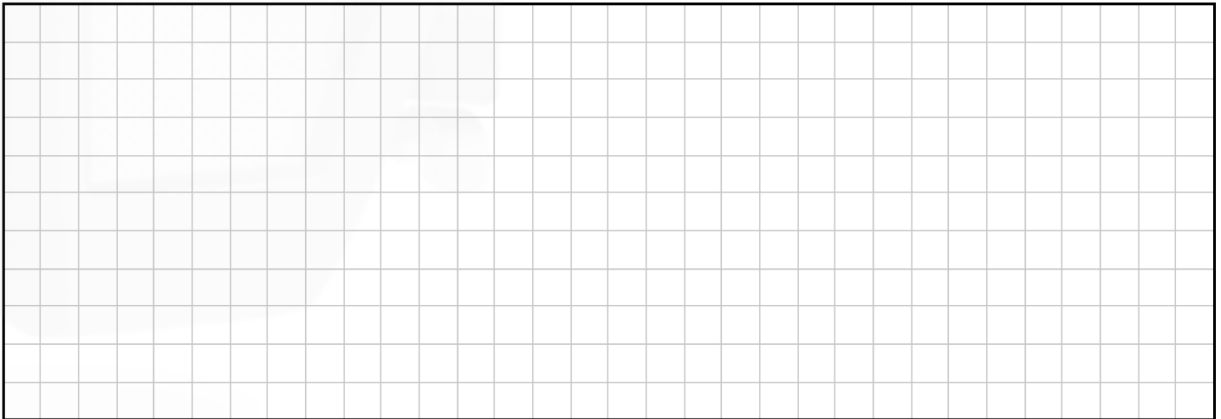
(ii) David is given three attempts to knock the coconut off its support. What is the probability that he knocks the coconut for the first time, on his **third attempt**? Give your answer correct to three decimal places.

(a) (i)	$1 - 0.34$ $= 0.66$	Scale 10C(0, 3, 7, 10) <i>Low Partial Credit:</i> Work of merit • Mention of 1 <i>High Partial Credit</i> Significant work • Writes $1 - 0.34$ <i>Full Credit:</i> • Correct answer without supporting work
(a) (ii)	$(0.66)(0.66)(0.34)$ $= 0.148104$ $= 0.148$	Scale 10C (0, 3, 7, 10) <i>Low Partial Credit:</i> Work of merit • Fail, Fail, Success <i>High Partial Credit:</i> Significant work • Correct answer indicated but not calculated $(0.66) \times (0.66) \times (0.34)$ <i>Full Credit:</i> • Correct answer without supporting work

- (b)** In a large population 1 in 8 of the people play tennis.
- (i)** Four people are chosen at random from the population.
What is the probability that the fourth person chosen is the only one to play tennis?



- (ii)** Three people are chosen at random from the population.
What is the probability that exactly two of them play tennis?



(b) (i)	$\frac{7}{8} \times \frac{7}{8} \times \frac{7}{8} \times \frac{1}{8} = \frac{343}{4096} \text{ or } 0.08374$	Scale 5C (0, 3, 4, 5) <i>Low Partial Credit:</i> Work of merit. - Probability as $\frac{1}{8}$ or $\frac{7}{8}$ <i>High Partial Credit:</i> Significant work $\frac{7}{8} \times \frac{7}{8} \times \frac{7}{8} \times \frac{1}{8}$ and stops Note: Treat $\frac{1}{8} \times \frac{1}{8} \times \frac{1}{8} \times \frac{7}{8}$ and continues correctly as <i>Full Credit</i> (–1) <i>Full Credit:</i> - Correct answer without supporting work
(b) (ii)	$3 \left(\frac{1}{8} \times \frac{1}{8} \times \frac{7}{8} \right) = \frac{21}{512} \text{ or } 0.04101$	Scale 5C (0, 3, 4, 5) <i>Low Partial Credit:</i> Work of merit $\frac{1}{8} \times \frac{7}{8}$ or $\frac{1}{8} \times \frac{1}{8}$ in any way <i>High Partial Credit:</i> Significant work $3 \left(\frac{1}{8} \times \frac{1}{8} \times \frac{7}{8} \right)$ or $\left(\frac{1}{8} \times \frac{1}{8} \times \frac{7}{8} \right) = \frac{7}{512}$ <i>Full Credit:</i> - Correct answer without supporting work

Question 1

(25 marks)

A business has 28 employees.
Their ages, in years, are given below.

32	41	57	64	19	21	35
18	43	54	63	65	33	22
39	58	18	42	20	34	21
49	33	55	34	57	43	63

1						
2						
3						
4						
5						
6						

KEY: $1 \mid 9 = 19$ years of age.

- (a)** Complete the stem-and-leaf diagram, showing the ages of all 28 employees.

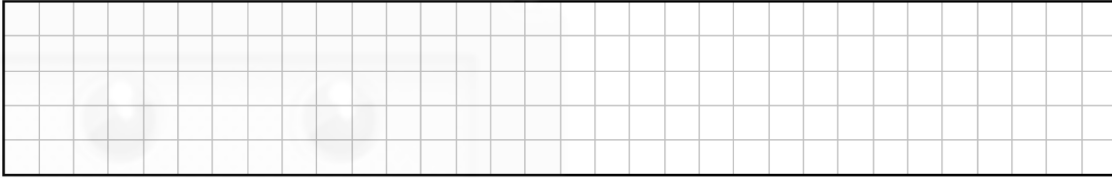
[illegible]

- (b)** Find the percentage of employees who are older than 40 years of age.

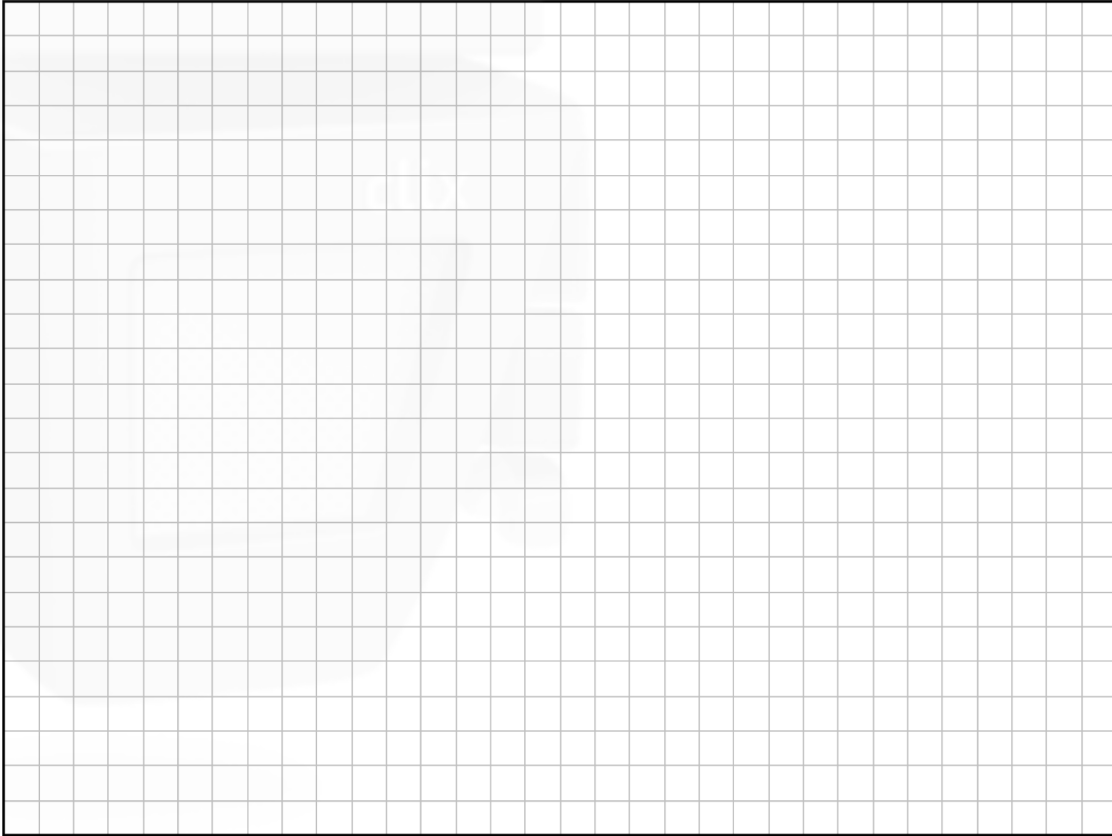
A blank sheet of graph paper with a grid pattern. The grid consists of small squares formed by thin gray lines. There are 20 columns and 15 rows of squares. A thicker black border runs along the top and left edges of the page.

(c) One employee is chosen at random on a day when all employees are present at work.

(i) Find the probability that the employee is a teenager (<20 years of age).



(ii) Find the probability that the employee chosen is a person in their thirties whose age is even or a person in their forties whose age is odd.



(a)	<table><tr><td>1</td><td>8</td><td>9</td><td>9</td><td></td><td></td><td></td><td></td></tr><tr><td>2</td><td>0</td><td>1</td><td>1</td><td>2</td><td></td><td></td><td></td></tr><tr><td>3</td><td>2</td><td>3</td><td>3</td><td>4</td><td>4</td><td>5</td><td>9</td></tr><tr><td>4</td><td>1</td><td>2</td><td>3</td><td>3</td><td>9</td><td></td><td></td></tr><tr><td>5</td><td>4</td><td>5</td><td>7</td><td>7</td><td>8</td><td></td><td></td></tr><tr><td>6</td><td>3</td><td>3</td><td>4</td><td>5</td><td></td><td></td><td></td></tr><tr><td colspan="8">KEY: 1 9 = 19 years of age</td></tr></table>	1	8	9	9					2	0	1	1	2				3	2	3	3	4	4	5	9	4	1	2	3	3	9			5	4	5	7	7	8			6	3	3	4	5				KEY: 1 9 = 19 years of age								Scale 10C (0, 4, 6, 10) <i>Low Partial Credit:</i> • Attempt at ordering ages • One correct entry <i>High Partial Credit:</i> • 15 correct entries <i>Full Credit:</i> • Correct answer without work
1	8	9	9																																																							
2	0	1	1	2																																																						
3	2	3	3	4	4	5	9																																																			
4	1	2	3	3	9																																																					
5	4	5	7	7	8																																																					
6	3	3	4	5																																																						
KEY: 1 9 = 19 years of age																																																										
(b)	$\frac{14}{28}$ $= 50\%$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> • Work of merit • Mention of 14 or 28 • Reference to $\times$ by 100 <i>High Partial Credit:</i> • Answer given as $\frac{14}{28}$ or equivalent <i>Full Credit:</i> • Correct answer without work																																																								

(c)(i)	$\frac{3}{28}$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> • Work of merit <i>High Partial Credit:</i> • $\frac{x}{28}, (x \neq 3, x \leq 28)$ • $\frac{3}{x}, (x \neq 28, x \geq 3)$ <i>Full Credit:</i> • Correct answer without work <i>Zero Credit:</i> • Probability > 1, without work
(c)(ii)	$= \frac{3}{28} + \frac{4}{28}$ $= \frac{7}{28}$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> • Work of merit <i>High Partial Credit:</i> • Both probabilities calculated correctly <i>Full Credit:</i> • Correct answer without work <i>Zero Credit:</i> • Probability > 1, without work

Question 1

(25 marks)

- (a) An experiment consists of throwing two fair, standard, six-sided dice and recording the sum of the two numbers thrown. Some of the totals are shown in the table.

- (i)** Complete the table.

- (ii) Find the probability of getting a total of 7 **or** 11.

[illegible]

		Die 1					
		1	2	3	4	5	6
Die 2	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6			9	10
	5	6		8	9		11
	6		8	9	10	11	

- (iii)** Find the probability of getting a total which is a prime number.

[illegible]

- (b) A car distributor sells Ford cars and Renault cars. It has 30 cars for sale on a particular day; 18 are Ford cars and 12 are Renault cars. 7 of the Ford cars are red and 4 of the Renault cars are red. One of the 30 cars is chosen at random. What is the probability that the car chosen is a Ford car **or** a car which is **not** red?

A full-page view of a blank sheet of graph paper. The grid consists of thin, light gray horizontal and vertical lines forming small squares across the entire page. There are no margins, text, or other markings on the paper.

Q1	Model Solution – 25 Marks	Marking Notes																																																									
(a) (i)	<table><tr><td colspan="2" rowspan="2"></td><td colspan="6">Die 1</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td rowspan="6">Die 2</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td></tr><tr><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td></tr></table>			Die 1						1	2	3	4	5	6	Die 2	1	2	3	4	5	6	7	2	3	4	5	6	7	8	3	4	5	6	7	8	9	4	5	6	7	8	9	10	5	6	7	8	9	10	11	6	7	8	9	10	11	12	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> - One correct entry. - Work of merit. <i>High Partial Credit:</i> - Four correct entries. <i>Full Credit:</i> - Correct answer without work. <i>Zero Credit:</i> - Correct answers outside box with no work.
				Die 1																																																							
		1	2	3	4	5	6																																																				
Die 2	1	2	3	4	5	6	7																																																				
	2	3	4	5	6	7	8																																																				
	3	4	5	6	7	8	9																																																				
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	5	6	7	8	9	10	11																																																				
	6	7	8	9	10	11	12																																																				
(a) (ii)	$\frac{6 + 2}{36} = \frac{8}{36} \text{ or } \frac{2}{9}$	Scale 5B (0, 2, 5) <i>Partial Credit:</i> - Work of merit. <i>Full Credit:</i> - Correct answer without work. <i>Zero Credit:</i> - Probability > 1, without work.																																																									
(a) (iii)	$\frac{1 + 2 + 4 + 6 + 2}{36} = \frac{15}{36} \text{ or } \frac{5}{12}$	Scale 5B (0, 2, 5) <i>Partial Credit:</i> - Prime Number identified. - Work of merit. <i>Full Credit:</i> - Correct answer without work. <i>Zero Credit:</i> - Probability > 1, without work.																																																									

(b)	$\frac{18 + 8}{30} = \frac{26}{30} = \frac{13}{15}$ Or $\frac{30 - 4}{30} = \frac{26}{30} = \frac{13}{15}$ Or $\frac{18 + (11 + 8) - 11}{30} = \frac{26}{30}$	Scale 10C (0, 3, 7, 10) <i>Low Partial Credit:</i> - Any relevant work towards creating at least one correct probability. (18 + 8, 30 – 4, 18 + 19 -11) <i>High Partial Credit:</i> - Significant work towards answer. <i>Full Credit:</i> - Correct answer without work. <i>Zero Credit:</i> - Probability > 1, without work.
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Question 7

(60 marks)

The table below shows the total rainfall, in millimetres, and the total sunshine, in hours, at Valentia, County Kerry, during the month of June from 2001 to 2010.

Total rainfall and total sunshine at Valentia in June										
Year	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
Total Rainfall (mm)	72	133	155	101	94	47	149	134	94	84
Total Sunshine (hours)	169	124	180	173	173	239	159	168	228	205

(Source: Met Éireann)

- (a) Based on the data in the table above write down:
- (i) the range of the rainfall data

(ii) the year with the highest June rainfall

(iii) the year with the least June sunshine
- (b) Based on the data in the table, write down the year with the best June weather and give a reason for your answer.

Answer:

Reason:

- (c) Write the **rainfall data** in increasing order and hence find the median of the rainfall.

- (d) (i) Find the mean number of **sunshine hours** for June in Valentia between 2001 and 2010.

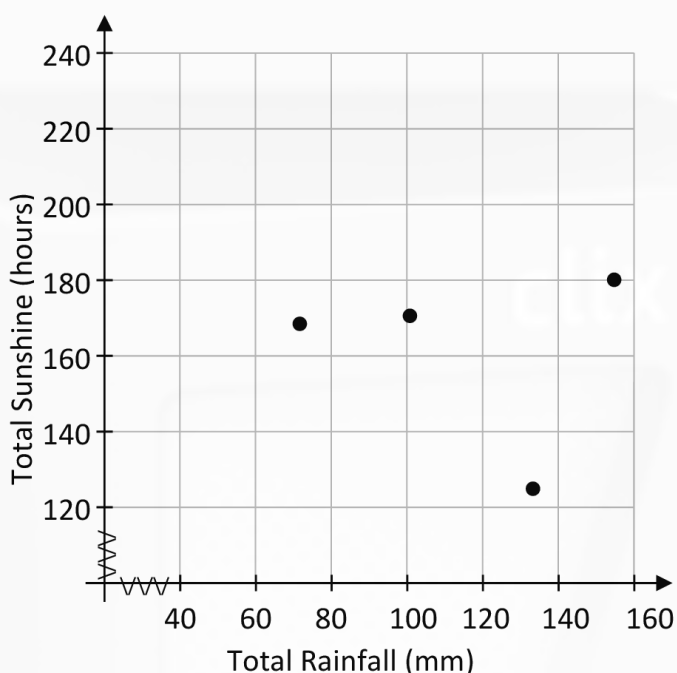
- (ii) For what years was the **sunshine data** within 5% of the mean number of sunshine hours in Valentia?

[illegible]

- (e) Find the standard deviation of the **rainfall data**, in mm, correct to 1 decimal place.

- (f) Part of a scatterplot of the data in the table is shown below. The first four data points are plotted.

- (i) Complete the scatterplot.



- (ii)** One of the numbers in the table on the right is the correlation coefficient for the data above, correct to 1 decimal place. Based on the scatterplot, select the number that you think most accurately reflects this data. Explain your choice.

	Tick one box
0.6	
0.1	
-0.1	
-0.6	

Explanation:

Explanation:

Section B

Q7	Model Solution – 60 Marks	Marking Notes
(a)	(i) Range= $155 - 47 = 108$ mm (ii) Highest June rainfall = 2003 (iii) Least sunshine 2002	Scale 10C (0, 3, 7, 10) <i>Low Partial Credit:</i> - Any work of merit e.g. (a)(i): [155 and/or 47] (a)(ii): [155] (a)(iii): [124] <i>High Partial Credit:</i> - Two correct answers. - Significant work in all answers.
(b)	<i>Example</i> 2006 is the year with the best weather. It had the least rainfall and /or the highest number of hours of sunshine. Note: Other equally valid choices are possible if accompanied by a reasonable reason.	Scale 5B (0, 2, 5) <i>Partial Credit:</i> - Mention of year without reference to rainfall or sunshine. <i>Full Credit:</i> - Mention of year with reference to rainfall and/or sunshine. <i>Zero Credit:</i> - A reason with no year identified.
(c)	47, 72, 84, 94, 94, 101, 133, 134, 149, 155. Median = $\frac{94+101}{2} = 97.5$ mm.	Scale 10C (0, 3, 7, 10) <i>Low Partial Credit:</i> - Effort at ordering data. - Median calculated as $\left(\frac{94+47}{2}\right) = 70.5$. - Consistent Median from an incorrect list (without work). <i>High Partial Credit:</i> - All Data ordered. - Correct Median without list. - Correct Median with Sunshine data [173]. <i>Full Credit:</i> - Correct list and correct Median without work. <i>Misreading:</i> - Works with a decreasing list.

(d) (i)	$\text{Mean} = \frac{\sum_1^{10} \text{Sunshine Data}}{10}$ $= \frac{1818}{10}$ $= 181.8$	Scale 10C (0, 3, 7, 10) <i>Low Partial Credit:</i> • Effort at adding data. • Division by 10. <i>High Partial Credit:</i> • Formula fully substituted without calculation. • Gets mean of the rainfall [=106.3]. <i>Full Credit:</i> • Correct answer without work.
(d) (ii)	$5\% \text{ of } 181.8 = 9.09$ $+5\% = 190.89 \text{ hours}$ $-5\% = 172.71 \text{ hours}$ Years within: 2003, 2004, 2005	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> • Any work of merit e.g. Identifies One year (without work). • Finds 5% of mean [Answer (d)(i)]. <i>High Partial Credit:</i> • Two of the years identified (with work). • Establishes [172.71, 190.89] or consistent. • Correct answers without work.
(e)	Rainfall $\sigma = 33.46057381$ $\sigma = 33.5$	Scale 5B (0, 2, 5) <i>Partial Credit:</i> • Mean calculated as 106.3. • Calculates σ for sunshine data [32.1]. • Note: • Answer as 33, or 33.46 merits F* .

(f)
(i)

Plots the remaining 6 pieces of data below.

Scale 10C (0, 3, 7, 10)

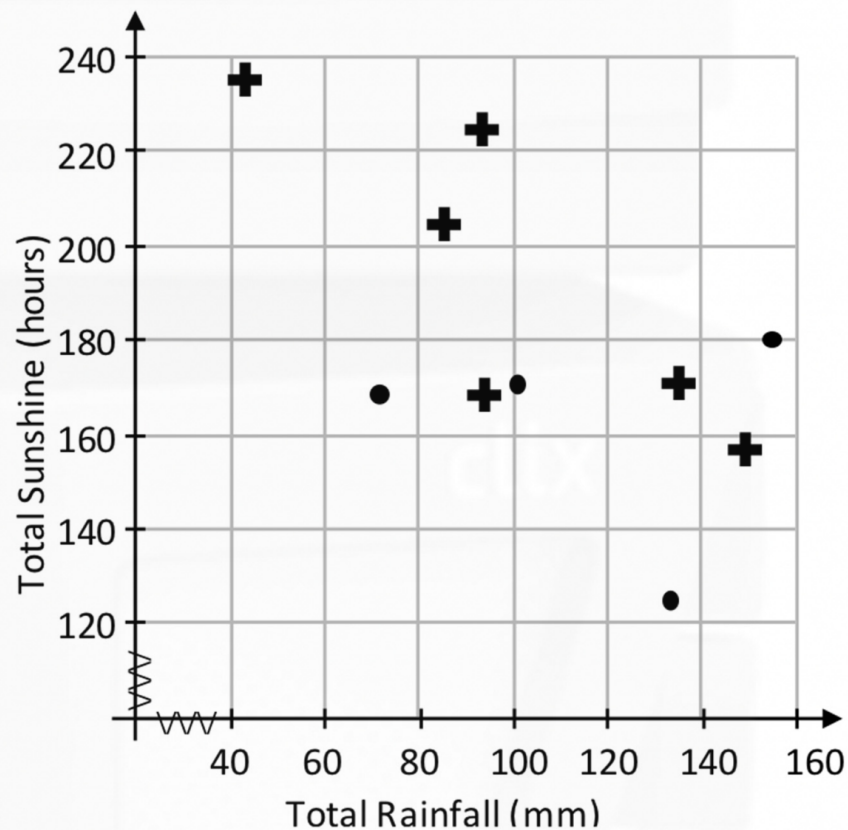
Low Partial Credit:

- One correct plot.

High Partial Credit:

- Four correct plots.

Tolerance: Within the correct box.



(f)
(ii)

	Tick one box
0·6	
0·1	
−0·1	
−0·6	✓

Explanation: Correlation is negative and moderate or strong.

Scale 5A (0, 5)

Full Credit:

- Correct box ticked and explanation is either:
Correlation is negative and strong.
or
Correlation is negative and moderate.

Zero Credit:

- All other answers.

